Topic V: Denoising

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We now turn to the problem of *denoising* (also called *prediction* or *estimation* in the statistical literature, and *filtering* in the signal processing literature). The goal is to recover the signal components of observed signal-plus-noise vectors.

1 The Wiener filter

To begin with, we again consider the random signal-plus-noise vector $Y = X + \varepsilon$, where $\varepsilon \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}_p)$; our goal is to estimate X from Y. Denote by $\boldsymbol{\mu} = \mathbb{E}[X] = \mathbb{E}[Y]$ the mean of X and Y (which are the same, since $\mathbb{E}[\varepsilon] = \mathbf{0}$. Also denote by $\mathbf{\Sigma}_x = \mathbb{E}[(X - \boldsymbol{\mu})(X - \boldsymbol{\mu})^T]$ the covariance of X, and by $\mathbf{\Sigma}_y = \mathbb{E}[(Y - \boldsymbol{\mu})(Y - \boldsymbol{\mu})^T] = \mathbf{\Sigma}_x + \sigma^2 \mathbf{I}_p$ the covariance of Y.

We will consider the class of *affine linear* denoisers, which are of the form

$$\widehat{X} = \mathbf{A}Y + \mathbf{b}, \tag{1}$$

where $\mathbf{A} \in \mathbb{R}^{p \times p}$ and $\mathbf{b} \in \mathbb{R}^p$ are deterministic. The mean squared error between $\widehat{X}$ and X is given by:

$$\mathrm{MSE}(\mathbf{A}, \mathbf{b}) = \mathbb{E}\|\mathbf{A}Y + \mathbf{b} - X\|^2, \tag{2}$$

where the expectation is taken over the randomness in both X and ε. We have the following result:

Proposition 1.1. *The mean squared error* $\mathrm{MSE}(\mathbf{A}, \mathbf{b})$ *is minimized by taking*

$$\mathbf{A} = \mathbf{\Sigma}_x(\mathbf{\Sigma}_x + \sigma^2 \mathbf{I}_p)^{-1} \tag{3}$$

and

$$\mathbf{b} = (\mathbf{I}_p - \mathbf{A})\boldsymbol{\mu}. \tag{4}$$

Remark 1. Note that the matrix $\mathbf{\Sigma}_x + \sigma^2 \mathbf{I}_p$ is invertible, since $\mathbf{\Sigma}_x$ is PSD.

By the proposition, the optimal affine linear denoiser is of the form

$$\widehat{X} = \mathbf{\Sigma}_x(\mathbf{\Sigma}_x + \sigma^2 \mathbf{I}_p)^{-1}(Y - \boldsymbol{\mu}) + \boldsymbol{\mu}. \tag{5}$$

The optimal affine linear denoiser given by equation (5) is known as the *Wiener filter*. Before proving the optimality of the Wiener filter, we state some preliminary lemmas.

Lemma 1.2. *Let W be random vector in $\mathbb{R}^p$ with mean zero and covariance $\mathbf{\Sigma}_w$, and let $\mathbf{B}$ be any p-by-p matrix. Then*

$$\mathbb{E}\langle \mathbf{B}W, W \rangle = \mathrm{tr}(\mathbf{B}^T \mathbf{\Sigma}_w). \tag{6}$$

Proof of Lemma 1.2. Since $\mathbb{E}[W] = \mathbf{0}$, $(\mathbf{\Sigma}_w)_{ij} = \mathbb{E}[w_i w_j]$. Consequently,

$$\begin{aligned}
\mathbb{E}\langle \mathbf{B}W, W \rangle &= \mathbb{E}\sum_{ij} b_{ij} w_i w_j \\
&= \sum_{ij} b_{ij} \mathbb{E}[w_i w_j] \\
&= \sum_{ij} b_{ij} (\mathbf{\Sigma}_w)_{ij} \\
&= \langle \mathbf{B}, \mathbf{\Sigma}_w \rangle \\
&= \mathrm{tr}(\mathbf{B}^T \mathbf{\Sigma}_w),
\end{aligned} \tag{7}$$

which is the desired result. □

Lemma 1.3. *Let* $\mathbf{B}$ *and* $\mathbf{C}$ *be any* p*-by-*p *matrices. Then*

$$\nabla_{\mathbf{B}} \text{tr}(\mathbf{B}\mathbf{C}\mathbf{B}^T) = \mathbf{B}(\mathbf{C} + \mathbf{C}^T). \tag{8}$$

Proof of Lemma 1.3. Fix indices j and m between 1 and p. Observe that

$$\begin{aligned}\text{tr}(\mathbf{B}\mathbf{C}\mathbf{B}^T) &= \sum_{i;k;\ell} b_{ik} b_{i\ell} c_{k\ell} \\ &= \sum_{i} b_{im} b_{im} c_{mm} + \sum_{i;k\neq m} b_{ik} b_{im} c_{km} + \sum_{i;\ell\neq m} b_{im} b_{i\ell} c_{m\ell} + \sum_{i;k\neq m;\ell\neq m} b_{ik} b_{i\ell} c_{k\ell}.\end{aligned} \tag{9}$$

Consequently,

$$\begin{aligned}\frac{\partial}{\partial b_{jm}} \text{tr}(\mathbf{B}\mathbf{C}\mathbf{B}^T) &= \frac{\partial}{\partial b_{jm}} \left[\sum_{i} b_{im} b_{im} c_{mm} + \sum_{i;k\neq m} b_{ik} b_{im} c_{km} + \sum_{i;\ell\neq m} b_{im} b_{i\ell} c_{m\ell} + \sum_{i;k\neq m;\ell\neq m} b_{ik} b_{i\ell} c_{k\ell} \right] \\ &= 2 b_{jm} c_{mm} + \sum_{k\neq m} b_{jk} c_{km} + \sum_{\ell\neq m} b_{j\ell} c_{m\ell} \\ &= \sum_{k} b_{jk} c_{km} + \sum_{\ell} b_{j\ell} c_{m\ell} \\ &= (\mathbf{B}\mathbf{C})_{jm} + (\mathbf{B}\mathbf{C}^T)_{jm},\end{aligned} \tag{10}$$

which yields the result. $\square$

Proof of Proposition 1.1. First, we find the optimal $\mathbf{b}$ in terms of $\mathbf{A}$. We write:

$$\begin{aligned}\text{MSE} &= \mathbb{E}\|\mathbf{A}Y + \mathbf{b} - X\|^2 \\ &= \mathbb{E}\|\mathbf{A}Y - X\|^2 + \mathbb{E}\|\mathbf{b}\|^2 + 2\mathbb{E}\langle \mathbf{A}Y - X, \mathbf{b}\rangle \\ &= \mathbb{E}\|\mathbf{A}Y - X\|^2 + \|\mathbf{b}\|^2 + 2\langle (\mathbf{A} - \mathbf{I}_p)\boldsymbol{\mu}, \mathbf{b}\rangle,\end{aligned} \tag{11}$$

and consequently

$$\nabla_{\mathbf{b}} \text{MSE} = 2\mathbf{b} + 2(\mathbf{A} - \mathbf{I}_p)\boldsymbol{\mu} \tag{12}$$

Solving $\nabla_{\mathbf{b}} \text{MSE}(\mathbf{A}, \mathbf{b}) = \mathbf{0}$ then yields

$$\mathbf{b} = (\mathbf{I}_p - \mathbf{A})\boldsymbol{\mu}, \tag{13}$$

which is the desired expression. We therefore rewrite the objective as a function of $\mathbf{A}$ alone:

$$\text{MSE} = \mathbb{E}\|\mathbf{A}(Y - \boldsymbol{\mu}) - (X - \boldsymbol{\mu})\|^2. \tag{14}$$

To keep the notation simpler, we will assume in the following computation that $\boldsymbol{\mu} = \mathbf{0}$. Write:

$$\begin{aligned}\|\mathbf{A}Y - X\|^2 &= \|\mathbf{A}Y\|^2 + \|X\|^2 - 2\langle \mathbf{A}Y, X\rangle \\ &= \|\mathbf{A}Y\|^2 + \|X\|^2 - 2\langle \mathbf{A}X, X\rangle - 2\langle \mathbf{A}\varepsilon, X\rangle, \\ &= \langle \mathbf{A}^T\mathbf{A}Y, Y\rangle + \langle X, X\rangle - 2\langle \mathbf{A}X, X\rangle - 2\langle \mathbf{A}\varepsilon, X\rangle,\end{aligned} \tag{15}$$

and taking expectations and using Lemma 1.2,

$$\begin{aligned}\text{MSE} &= \mathbb{E}\|\mathbf{A}Y - X\|^2 \\ &= \mathbb{E}\langle \mathbf{A}^T\mathbf{A}Y, Y\rangle + \mathbb{E}\langle X, X\rangle - 2\mathbb{E}\langle \mathbf{A}X, X\rangle - 2\mathbb{E}\langle \mathbf{A}\varepsilon, X\rangle \\ &= \mathbb{E}\langle \mathbf{A}^T\mathbf{A}Y, Y\rangle + \mathbb{E}\langle X, X\rangle - 2\mathbb{E}\langle \mathbf{A}X, X\rangle \\ &= \text{tr}(\boldsymbol{\Sigma}_y\mathbf{A}^T\mathbf{A}) - 2\text{tr}(\boldsymbol{\Sigma}_x\mathbf{A}^T) + \text{tr}(\boldsymbol{\Sigma}_x). \\ &= \text{tr}(\mathbf{A}\boldsymbol{\Sigma}_y\mathbf{A}^T) - 2\text{tr}(\boldsymbol{\Sigma}_x\mathbf{A}^T) + \text{tr}(\boldsymbol{\Sigma}_x).\end{aligned} \tag{16}$$

By Lemma 1.3,

$$\nabla_{\mathbf{A}} \mathrm{MSE} = 2\mathbf{A}\boldsymbol{\Sigma}_y - 2\boldsymbol{\Sigma}_x, \tag{17}$$

and setting this equal to $\mathbf{0}$ and solving for $\mathbf{A}$ yields

$$\begin{aligned} \mathbf{A} &= \boldsymbol{\Sigma}_x \boldsymbol{\Sigma}_y^{-1} \\ &= \boldsymbol{\Sigma}_x (\boldsymbol{\Sigma}_x + \sigma^2 \mathbf{I}_p)^{-1}, \end{aligned} \tag{18}$$

which is the desired result. □

In fact, the Wiener filter also satisfies a stronger optimality criterion when X is assumed to be Gaussian. Take *any* estimator $f(Y)$ of X given Y (not necessarily linear). Write the MSE of f as

$$\begin{aligned} \mathrm{MSE}(f) &= \mathbb{E}_{X,Y} \|f(Y) - X\|^2 \\ &= \mathbb{E}_Y \mathbb{E}_{X|Y} \|f(Y) - X\|^2, \end{aligned} \tag{19}$$

and the inner expectation $\mathbb{E}_{X|Y} \|f(Y) - X\|^2$ is minimized by taking

$$f(Y) = \mathbb{E}[X|Y]. \tag{20}$$

To compute $\mathbb{E}[X|Y]$, we must evaluate the posterior distribution $p(X|Y)$, and find its mean.

Proposition 1.4. *Suppose $X \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma}_x)$. Then the Wiener filter, given by* (5)*, is the posterior mean $\mathbb{E}[X|Y]$; that is, it minimizes the $\mathbb{E}\|\widehat{X} - X\|^2$ over* all *estimators $\widehat{X}$.*

Proof. The posterior distribution of X given Y is:

$$\begin{aligned} p(X|Y) &= \frac{p(Y|X)p(X)}{p(Y)} \\ &\propto \exp\left\{-\frac{1}{2}\|Y - X\|^2/\sigma^2\right\} \exp\left\{-\frac{1}{2}(X - \boldsymbol{\mu})^T \boldsymbol{\Sigma}_x^{-1} (X - \boldsymbol{\mu})\right\} \\ &\propto \exp\left\{-\frac{1}{2}\left[\|Y - X\|^2/\sigma^2 + (X - \boldsymbol{\mu})^T \boldsymbol{\Sigma}_x^{-1} (X - \boldsymbol{\mu})\right]\right\}. \end{aligned} \tag{21}$$

Then the expression in square brackets may be rewritten as follows (where C denotes a constant not depending on X):

$$\begin{aligned} \|Y - X\|^2/\sigma^2 + (X - \boldsymbol{\mu})^T \boldsymbol{\Sigma}_x^{-1} (X - \boldsymbol{\mu}) &= \frac{1}{\sigma^2} Y^T Y + \frac{1}{\sigma^2} X^T X - \frac{2}{\sigma^2} Y^T X + X^T \boldsymbol{\Sigma}_x^{-1} X + \boldsymbol{\mu}^T \boldsymbol{\Sigma}_x^{-1} \boldsymbol{\mu} - 2X^T \boldsymbol{\Sigma}^{-1} \boldsymbol{\mu} \\ &= X^T (\sigma^{-2} \mathbf{I}_p + \boldsymbol{\Sigma}_x^{-1}) X - 2(\sigma^{-2} Y + \boldsymbol{\Sigma}_x^{-1} \boldsymbol{\mu})^T X + C. \end{aligned} \tag{22}$$

Letting

$$\mathbf{A} = \sigma^{-2} \mathbf{I}_p + \boldsymbol{\Sigma}_x^{-1} \tag{23}$$

and

$$\mathbf{b} = \sigma^{-2} Y + \boldsymbol{\Sigma}_x^{-1} \boldsymbol{\mu}, \tag{24}$$

we may complete the square and write:

$$\begin{aligned} X^T (\sigma^{-2} \mathbf{I}_p + \boldsymbol{\Sigma}_x^{-1}) X - 2(\sigma^{-2} Y + \boldsymbol{\Sigma}_x^{-1} \boldsymbol{\mu})^T X &= X^T \mathbf{A} X - 2\mathbf{b}^T X \\ &= (X - \mathbf{A}^{-1}\mathbf{b})^T \mathbf{A} (X - \mathbf{A}^{-1}\mathbf{b}) + C, \end{aligned} \tag{25}$$

where again C does not depend on X, from which we see that the posterior distribution is of the form

$$p(X|Y) \propto \exp\left\{-\frac{1}{2}(X - \mathbf{A}^{-1}\mathbf{b})^T \mathbf{A} (X - \mathbf{A}^{-1}\mathbf{b})\right\}, \tag{26}$$

and hence $X|Y \sim \mathcal{N}(\mathbf{A}^{-1}\mathbf{b}, \mathbf{A}^{-1})$. Consequently, the posterior mean is given by

$$\begin{aligned}
\mathbb{E}[X|Y] &= \mathbf{A}^{-1}\mathbf{b} \\
&= (\sigma^{-2}\mathbf{I}_p + \mathbf{\Sigma}_x^{-1})^{-1}(\sigma^{-2}Y + \mathbf{\Sigma}_x^{-1}\boldsymbol{\mu}) \\
&= (\mathbf{I}_p + \sigma^2\mathbf{\Sigma}_x^{-1})^{-1}(Y + \sigma^2\mathbf{\Sigma}_x^{-1}\boldsymbol{\mu}) \\
&= (\mathbf{I}_p + \sigma^2\mathbf{\Sigma}_x^{-1})^{-1}(Y - \boldsymbol{\mu}) + (\mathbf{I}_p + \sigma^2\mathbf{\Sigma}_x^{-1})^{-1}(\boldsymbol{\mu} + \sigma^2\mathbf{\Sigma}_x^{-1}\boldsymbol{\mu}) \\
&= (\mathbf{I}_p + \sigma^2\mathbf{\Sigma}_x^{-1})^{-1}(Y - \boldsymbol{\mu}) + (\mathbf{I}_p + \sigma^2\mathbf{\Sigma}_x^{-1})^{-1}(\mathbf{I}_p + \sigma^2\mathbf{\Sigma}_x^{-1})\boldsymbol{\mu} \\
&= (\mathbf{I}_p + \sigma^2\mathbf{\Sigma}_x^{-1})^{-1}(Y - \boldsymbol{\mu}) + \boldsymbol{\mu} \\
&= \mathbf{\Sigma}_x(\mathbf{\Sigma}_x + \sigma^2\mathbf{I}_p)^{-1}(Y - \boldsymbol{\mu}) + \boldsymbol{\mu},
\end{aligned} \tag{27}$$

which is the Wiener filter. □

Remark 2. Another way of stating Proposition 1.4 is that Wiener filter is the *Bayes estimator* of X when $\mathbf{X}$ has the Gaussian prior $\mathcal{N}(\boldsymbol{\mu}, \mathbf{\Sigma}_x)$. Chapters 2 and 3 of [6] contains a brief but fairly comprehensive introduction to Bayes estimation and related notions in decision theory.

1.1 Estimating the Wiener filter

In practice, we typically do not know $\mathbf{\Sigma}_x$. In typical applications, it is easier to acquire many noise-only samples than signal-plus-noise samples, so we will treat the matrix σ^2 as *known* (and it is often not hard to estimate from signal-plus-noise samples, given some weak assumptions). However, the matrix $\mathbf{\Sigma}_x$ is *not* known a priori, so the Wiener filter $\widehat{X}$ cannot be applied. However, if we observe many independent observations $Y_1, \dots, Y_n$, as we have seen we can consistently estimate $\mathbf{\Sigma}_x$ as follows. First, form the sample covariance of Y,

$$\widehat{\mathbf{\Sigma}}_{y,n} = \frac{1}{n}\sum_{j=1}^{n}(Y_j - \widehat{\boldsymbol{\mu}}_n)(Y_j - \widehat{\boldsymbol{\mu}}_n)^T, \tag{28}$$

where

$$\widehat{\boldsymbol{\mu}}_n = \frac{1}{n}\sum_{j=1}^{n} Y_j \tag{29}$$

is the sample mean. As we know, as $n \to \infty$ the sample mean $\widehat{\boldsymbol{\mu}}_n$ converges almost surely to $\boldsymbol{\mu}$, and the sample covariance converges almost surely to $\mathbf{\Sigma}_y = \mathbf{\Sigma}_x + \sigma^2\mathbf{I}_p$. We then have only to subtract off the noise covariance $\sigma^2\mathbf{I}_p$, to produce an estimate of $\mathbf{\Sigma}_x$:

$$\widehat{\mathbf{\Sigma}}_{x,n} = \widehat{\mathbf{\Sigma}}_{y,n} - \sigma^2\mathbf{I}_p. \tag{30}$$

Consequently, to denoise the vectors $Y_1, \dots, Y_n$, we can use the predictor

$$\widehat{X}_j = \widehat{\mathbf{\Sigma}}_{x,n}(\widehat{\mathbf{\Sigma}}_{x,n} + \sigma^2\mathbf{I}_p)^{-1}(Y_j - \widehat{\boldsymbol{\mu}}_n) + \widehat{\boldsymbol{\mu}}_n, \tag{31}$$

for $1 \le j \le n$.

1.2 The Wiener filter and singular value shrinkage

There is a way of rewriting the predictor that is particularly convenient. To keep the notation and exposition simple, we will assume that $\boldsymbol{\mu} = \mathbf{0}$ is known, so that

$$\widehat{X}_j = \widehat{\mathbf{\Sigma}}_{x,n}(\widehat{\mathbf{\Sigma}}_{x,n} + \sigma^2\mathbf{I}_p)^{-1}Y_j. \tag{32}$$

Let $\widehat{\mathbf{X}} = [\widehat{X}_1, \dots, \widehat{X}_n]/\sqrt{n}$ and $\mathbf{Y} = [Y_1, \dots, Y_n]/\sqrt{n}$ (note the normalization by $\sqrt{n}$). Write the SVD of $\mathbf{Y}$ as follows:

$$\mathbf{Y} = \widehat{\mathbf{U}}\widehat{\mathbf{\Lambda}}^{1/2}\widehat{\mathbf{V}}^T \tag{33}$$

Then the left singular vectors $\widehat{\mathbf{U}}$ are the eigenvectors of $\widehat{\mathbf{\Sigma}}_y = \mathbf{Y}\mathbf{Y}^T$, and the singular values of $\mathbf{Y}$ are the square roots of the eigenvalues of $\widehat{\mathbf{\Sigma}}_y$. We can write

$$\widehat{\mathbf{X}} = \widehat{\mathbf{\Sigma}}_{x,n}(\widehat{\mathbf{\Sigma}}_{x,n} + \sigma^2\mathbf{I}_p)^{-1}\mathbf{Y}. \tag{34}$$

Plugging in the SVD gives $\mathbf{Y} = \widehat{\mathbf{U}}\widehat{\mathbf{\Lambda}}^{1/2}\widehat{\mathbf{V}}^T$ and the eigendecomposition

$$\widehat{\mathbf{\Sigma}}_{x,n}(\widehat{\mathbf{\Sigma}}_{x,n} + \sigma^2\mathbf{I}_p)^{-1} = \widehat{\mathbf{U}}(\widehat{\mathbf{\Lambda}} - \sigma^2\mathbf{I}_p)\widehat{\mathbf{\Lambda}}^{-1}\widehat{\mathbf{U}}^T \tag{35}$$

allows us to write $\widehat{\mathbf{X}}$ as follows:

$$\begin{aligned}\widehat{\mathbf{X}} &= \widehat{\mathbf{\Sigma}}_{x,n}(\widehat{\mathbf{\Sigma}}_{x,n} + \sigma^2\mathbf{I}_p)^{-1}\mathbf{Y} \\ &= \widehat{\mathbf{U}}(\widehat{\mathbf{\Lambda}} - \sigma^2\mathbf{I}_p)\widehat{\mathbf{\Lambda}}^{-1}\widehat{\mathbf{U}}^T\widehat{\mathbf{U}}\widehat{\mathbf{\Lambda}}^{1/2}\widehat{\mathbf{V}}^T \\ &= \widehat{\mathbf{U}}(\widehat{\mathbf{\Lambda}} - \sigma^2\mathbf{I}_p)\widehat{\mathbf{\Lambda}}^{-1/2}\widehat{\mathbf{V}}^T. \end{aligned} \tag{36}$$

Recalling that the SVD of $\mathbf{Y}$ is $\mathbf{Y} = \widehat{\mathbf{U}}\widehat{\Lambda}^{1/2}\widehat{\mathbf{V}}^T$, we see that $\widehat{\mathbf{X}}$ only acts on the singular values of $\mathbf{Y}$, replacing each $\sqrt{\widehat{\lambda}_j}$ by the value

$$\widehat{t}_j = \frac{\widehat{\lambda}_j - \sigma^2}{\sqrt{\widehat{\lambda}_j}}, \quad 1 \le j \le p. \tag{37}$$

These are smaller than the singular values $\sqrt{\widehat{\lambda}_j}$ of $\mathbf{Y}$, and hence $\widehat{\mathbf{X}}$ is referred to as a *singular value shrinkage* estimator.

As $n \to \infty$ the singular value shrinkage method with singular values $\widehat{t}_j$ converges to the Wiener filter, which is optimal in a certain sense. However, this still leaves open the question of whether singular value shrinkage retains optimality guarantees when both n and p are large, and if so, how to optimally shrink the singular values. We will address both questions in the next section.

2 Singular value shrinkage

In this section, we let $\mathbf{X}$ denote a general p-by-n signal matrix, without assuming its columns are iid vectors. We observe the matrix $\mathbf{Y} = \mathbf{X} + \mathbf{N}$, where the $p \cdot n$ entries of $\mathbf{N}$ are iid $\mathcal{N}(0, 1/n)$. We will study the problem of matrix denoising, i.e. estimating $\mathbf{X}$ from $\mathbf{Y}$.

2.1 Optimality of functions of the singular values

Let $\mathcal{X}$ denote some set of p-by-n matrices that is orthogonally invariant; that is, $\mathbf{X}$ is contained in $\mathcal{X}$ if and only if $\mathbf{QXP}$ is contained in $\mathcal{X}$, where $\mathbf{P}$ and $\mathbf{Q}$ are orthogonal. For example, $\mathcal{X}$ might be the set of all matrices with prescribed singular values; or the set of all matrices with Frobenius norm not exceeding a certain threshold; etc.

Take *any* denoiser $\eta(\mathbf{Y})$. The maximum risk of η over $\mathcal{X}$ is given by

$$R(\eta) = \sup_{\mathbf{X} \in \mathcal{X}} \mathbb{E}_{\mathbf{N}}\|\eta(\mathbf{Y}) - \mathbf{X}\|_{\mathrm{F}}^2 \tag{38}$$

where the expectation is over all realizations of the noise matrix $\mathbf{N} = [\varepsilon_1, \ldots, \varepsilon_n]$. Note that we do not average over $\mathbf{X}$; we are looking at the worst-case performance of η over all possible $\mathbf{X}$ in $\mathcal{X}$.

Following [2], we will show that whatever η is, the following method is at least as good:

$$\eta_0(\mathbf{Y}) = \mathbb{E}_{\widetilde{\mathbf{U}},\widetilde{\mathbf{V}}}[\widetilde{\mathbf{U}}\eta(\widetilde{\mathbf{U}}^T\mathbf{Y}\widetilde{\mathbf{V}})\widetilde{\mathbf{V}}^T] \tag{39}$$

where the expectation here is over $\widetilde{\mathbf{U}} \in O(p)$ and $\widetilde{\mathbf{V}} \in O(n)$ with Haar measure. Before proving this, observe the following:

by performing the change of variable $\mathbf{U}_0 = \widehat{\mathbf{U}}^T\widetilde{\mathbf{U}}$ and $\mathbf{V}_0 = \widehat{\mathbf{V}}^T\widetilde{\mathbf{V}}$ in the expectation defining η_0, we get:

$$\begin{aligned}
\eta_0(\mathbf{Y}) &= \mathbb{E}_{\widetilde{\mathbf{U}},\widetilde{\mathbf{V}}}[\widetilde{\mathbf{U}}\eta(\widetilde{\mathbf{U}}^T\mathbf{Y}\widetilde{\mathbf{V}})\widetilde{\mathbf{V}}^T] \\
&= \mathbb{E}_{\widetilde{\mathbf{U}},\widetilde{\mathbf{V}}}[\widetilde{\mathbf{U}}\eta(\widetilde{\mathbf{U}}^T\widehat{\mathbf{U}}\widehat{\mathbf{\Lambda}}^{1/2}\widehat{\mathbf{V}}^T\widetilde{\mathbf{V}})\widetilde{\mathbf{V}}^T] \\
&= \mathbb{E}_{\mathbf{U}_0,\mathbf{V}_0}[\widehat{\mathbf{U}}\mathbf{U}_0\eta(\mathbf{U}_0^T\widehat{\mathbf{\Lambda}}^{1/2}\mathbf{V}_0)\mathbf{V}_0^T\widehat{\mathbf{V}}^T] \\
&= \widehat{\mathbf{U}}\mathbb{E}_{\mathbf{U}_0,\mathbf{V}_0}[\mathbf{U}_0\eta(\mathbf{U}_0^T\widehat{\mathbf{\Lambda}}^{1/2}\mathbf{V}_0)\mathbf{V}_0^T]\widehat{\mathbf{V}}^T \\
&= \widehat{\mathbf{U}}\eta_0(\widehat{\mathbf{\Lambda}}^{1/2})\widehat{\mathbf{V}}^T.
\end{aligned} \tag{40}$$

Hence the optimal denoiser is of the form $\widehat{\mathbf{U}}\eta_0(\widehat{\mathbf{\Lambda}}^{1/2})\widehat{\mathbf{V}}^T$; that is, it only acts on the singular values of $\mathbf{Y}$.

We now show $R(\eta_0) \le R(\eta)$. Since the noise matrix is Gaussian, and hence orthogonally-invariant, for any orthogonal matrices $\widetilde{\mathbf{U}}$ and $\widetilde{\mathbf{V}}$, $\widetilde{\mathbf{U}}^T\mathbf{N}\widetilde{\mathbf{V}} \sim \mathbf{N}$. We write:

$$\begin{aligned}
\sup_{\mathbf{X}} \mathbb{E}_{\mathbf{N}}\|\eta(\mathbf{Y}) - \mathbf{X}\|_{\mathrm{F}}^2 &= \sup_{\mathbf{U},\mathbf{V},\mathbf{\Theta}} \mathbb{E}_{\mathbf{N}}\|\eta(\mathbf{Y}) - \mathbf{U}\mathbf{\Theta}\mathbf{V}^T\|_{\mathrm{F}}^2 \\
&= \sup_{\mathbf{U},\mathbf{V},\mathbf{\Theta},\widetilde{\mathbf{U}},\widetilde{\mathbf{V}}} \mathbb{E}_{\mathbf{N}}\|\eta(\widetilde{\mathbf{U}}^T\mathbf{Y}\widetilde{\mathbf{V}}) - \widetilde{\mathbf{U}}^T\mathbf{U}\mathbf{\Theta}\mathbf{V}^T\widetilde{\mathbf{V}}\|_{\mathrm{F}}^2 \\
&= \sup_{\mathbf{U},\mathbf{V},\mathbf{\Theta},\widetilde{\mathbf{U}},\widetilde{\mathbf{V}}} \mathbb{E}_{\mathbf{N}}\|\widetilde{\mathbf{U}}\eta(\widetilde{\mathbf{U}}^T\mathbf{Y}\widetilde{\mathbf{V}})\widetilde{\mathbf{V}}^T - \mathbf{U}\mathbf{\Theta}\mathbf{V}^T\|_{\mathrm{F}}^2 \\
&\ge \sup_{\mathbf{U},\mathbf{V},\mathbf{\Theta}} \mathbb{E}_{\mathbf{N}}\mathbb{E}_{\widetilde{\mathbf{U}},\widetilde{\mathbf{V}}}\|\widetilde{\mathbf{U}}\eta(\widetilde{\mathbf{U}}^T\mathbf{Y}\widetilde{\mathbf{V}})\widetilde{\mathbf{V}}^T - \mathbf{U}\mathbf{\Theta}\mathbf{V}^T\|_{\mathrm{F}}^2 \\
&\ge \sup_{\mathbf{U},\mathbf{V},\mathbf{\Theta}} \mathbb{E}_{\mathbf{N}}\|\mathbb{E}_{\widetilde{\mathbf{U}},\widetilde{\mathbf{V}}}[\widetilde{\mathbf{U}}\eta(\widetilde{\mathbf{U}}^T\mathbf{Y}\widetilde{\mathbf{V}})\widetilde{\mathbf{V}}^T] - \mathbf{U}\mathbf{\Theta}\mathbf{V}^T\|_{\mathrm{F}}^2 \\
&= \sup_{\mathbf{X}} \mathbb{E}_{\mathbf{N}}\|\eta_0(\mathbf{Y}) - \mathbf{X}\|_{\mathrm{F}}^2,
\end{aligned} \tag{41}$$

where the last inequality follows from Jensen's inequality.

2.2 Singular value shrinkage in the spiked covariance model

From the preceding, the optimal denoiser of $\mathbf{X}$ given $\mathbf{Y} = \mathbf{X} + \mathbf{N}$ is of the form $\eta(\mathbf{Y}) = \widehat{\mathbf{U}}\eta(\mathbf{\Lambda}^{1/2})\widehat{\mathbf{V}}^T$. That is, the optimal denoised matrix $\widehat{\mathbf{X}}$ is of the form

$$\widehat{\mathbf{X}} = \widehat{\mathbf{U}}\widehat{\mathbf{Q}}\widehat{\mathbf{V}}^T, \tag{42}$$

for some r-by-r matrix $\widehat{\mathbf{Q}}$, as yet undetermined. Let us consider the problem of finding the optimal such $\widehat{\mathbf{Q}}$ in the high-dimensional setting of the spiked model. That is, suppose $\mathbf{X}$ has rank r, with SVD

$$\mathbf{X} = \sum_{k=1}^{r} \theta_k \mathbf{u}_{k,n}\mathbf{v}_{k,n}^T = \mathbf{U}\Theta\mathbf{V}^T, \tag{43}$$

where $\theta_1 > \dots \theta_r > 0$ are fixed parameters. Suppose $\mathbf{N}$ is a matrix with iid $\mathcal{N}(0, 1/n)$ entries; and let $\mathbf{Y} = \mathbf{X} + \mathbf{N}$ have SVD

$$\mathbf{Y} = \sum_{k=1}^{\min\{p,n\}} \sigma_{k,n}\widehat{\mathbf{u}}_{k,n}\widehat{\mathbf{v}}_{k,n}^T. \tag{44}$$

In this setting, we will formulate the denoising problem as follows: fix a deterministic r-by-r matrix $\mathbf{Q}$, and consider the estimator

$$\widehat{\mathbf{X}}(\mathbf{Q}) = \widehat{\mathbf{U}}\mathbf{Q}\widehat{\mathbf{V}}^T, \tag{45}$$

where $\widehat{\mathbf{U}} = [\widehat{\mathbf{u}}_{1,n}, \dots, \widehat{\mathbf{u}}_{r,n}]$ and $\widehat{\mathbf{V}} = [\widehat{\mathbf{v}}_{1,n}, \dots, \widehat{\mathbf{v}}_{r,n}]$. Define the asymptotic mean squared error as

$$\mathrm{AMSE}(\mathbf{Q}) = \lim_{n\to\infty} \|\widehat{\mathbf{X}}(\mathbf{Q}) - \mathbf{X}\|_{\mathrm{F}}^2, \tag{46}$$

where the limit is taken as $p, n \to \infty$ and $p/n \to \gamma$. Though this limit is not a priori well-defined, we will show below that it exists almost surely for any fixed $\mathbf{Q}$. Our goal, then, is to find the optimal such $\mathbf{Q}$, and estimate it, when $\mathbf{Y} = \mathbf{X} + \mathbf{N}$ is drawn from the spiked model. Define the values

$$c_k = \begin{cases} (\theta_k^4 - \gamma)/(\theta_k^4 + \gamma\theta_k^2), & \text{if } \theta_k^2 > \sqrt{\gamma} \\ 0, & \text{otherwise} \end{cases}, \tag{47}$$

and

$$\underline{c}_k = \begin{cases} (\theta_k^4 - \gamma)/(\theta_k^4 + \theta_k^2), & \text{if } \theta_k^2 > \sqrt{\gamma} g \\ 0, & \text{otherwise} \end{cases}. \tag{48}$$

We have the following result:

Proposition 2.1. *For any matrix $\mathbf{Q}$, the* AMSE *defined by* (46) *is well-defined almost surely, and is minimized by*

$$\mathbf{Q}^* = \mathrm{diag}(q_1^*, \dots, q_r^*), \tag{49}$$

where

$$q_k^* = \theta_k c_k \underline{c}_k. \tag{50}$$

The resulting AMSE *at $\mathbf{Q}^*$ is equal to*

$$\mathrm{AMSE}(\mathbf{Q}^*) = \sum_{k=1}^{r} \theta_k^2 (1 - c_k^2 \underline{c}_k^2). \tag{51}$$

Proof. The error may be written as follows:

$$\begin{aligned} \|\widehat{\mathbf{X}}(\mathbf{Q}) - \mathbf{X}\|_{\mathrm{F}}^2 &= \|\widehat{\mathbf{X}}(\mathbf{Q})\|_{\mathrm{F}}^2 - 2\langle \widehat{\mathbf{X}}(\mathbf{Q}), \mathbf{X} \rangle + \|\mathbf{X}\|_{\mathrm{F}}^2 \\ &= \|\mathbf{Q}\|_{\mathrm{F}}^2 - 2\mathrm{tr}(\widehat{\mathbf{U}}\mathbf{Q}\widehat{\mathbf{V}}^T \mathbf{V}\mathbf{\Theta}\mathbf{U}^T) + \|\mathbf{X}\|_{\mathrm{F}}^2 \\ &= \|\mathbf{Q}\|_{\mathrm{F}}^2 - \mathrm{tr}(\mathbf{Q}\widehat{\mathbf{V}}^T \mathbf{V}\mathbf{\Theta}\mathbf{U}^T \widehat{\mathbf{U}}) + \|\mathbf{X}\|_{\mathrm{F}}^2. \end{aligned} \tag{52}$$

Now, the matrix $\widehat{\mathbf{V}}^T \mathbf{V}\mathbf{\Theta}\mathbf{U}^T \widehat{\mathbf{U}}$ converges almost surely:

$$\widehat{\mathbf{V}}^T \mathbf{V}\mathbf{\Theta}\mathbf{U}^T \widehat{\mathbf{U}} \to \mathbf{D} \equiv \mathrm{diag}(\theta_1 c_1 \underline{c}_1, \dots, \theta_r c_r \underline{c}_r), \tag{53}$$

and so

$$\mathrm{AMSE}(\mathbf{Q}) = \|\mathbf{Q}\|_{\mathrm{F}}^2 - 2\mathrm{tr}(\mathbf{Q}\mathbf{D}) + \|\mathbf{X}\|_{\mathrm{F}}^2. \tag{54}$$

Since $\mathbf{D}$ is diagonal, the minimizer of AMSE($\mathbf{Q}$) must also be diagonal, since any non-zero off-diagonal entries would only enlarge $\|\mathbf{Q}\|_{\mathrm{F}}^2$. Writing $\mathbf{Q} = \mathrm{diag}(q_1, \dots, q_r)$ and dropping the term $\|\mathbf{X}\|_{\mathrm{F}}^2 = \sum_{j=1}^r \theta_j^2$ which is independent of $\mathbf{Q}$, we must minimize:

$$\|\mathbf{Q}\|_{\mathrm{F}}^2 - 2\mathrm{tr}(\mathbf{Q}\mathbf{D}) = \sum_{k=1}^{r} \left(q_k^2 - 2\theta_k c_k \underline{c}_k q_k \right), \tag{55}$$

which is minimized at the values

$$q_k^* = \theta_k c_k \underline{c}_k. \tag{56}$$

Plugging $\mathbf{Q}^* = \mathrm{diag}(q_1^*, \dots, q_r^*)$ into AMSE($\mathbf{Q}$) then yields

$$\begin{aligned} \mathrm{AMSE}(\mathbf{Q}^*) &= \|\mathbf{Q}^*\|_{\mathrm{F}}^2 - 2\mathrm{tr}(\mathbf{Q}^*\mathbf{D}) + \|\mathbf{X}\|_{\mathrm{F}}^2 \\ &= \sum_{k=1}^{r} ((q_k^*)^2 - 2\theta_k c_k \underline{c}_k q_k^* + \theta_k^2) \\ &= \sum_{k=1}^{r} ((\theta_k c_k \underline{c}_k)^2 - 2(\theta_k c_k \underline{c}_k)^2 + \theta_k^2) \\ &= \sum_{k=1}^{r} \theta_k^2 (1 - c_k^2 \underline{c}_k^2), \end{aligned} \tag{57}$$

as claimed. □

Remark 3. Note that the parameters θ_k, c_k and $\underline{c}_k$ can all be consistently estimated in the limit $n \to \infty$ as functions of the singular values σ_k of $\mathbf{Y}$. Therefore, the optimal $q_1^*, \dots, q_r^*$ may also be estimated.

The optimal shrinker we derived here may be found in [5], and in a slightly different form in [3]; the latter paper also provides a convenient framework for deriving optimal shrinkers with respect to other loss functions, beyond the Frobenius loss we have considered here. The related problem of eigenvalue shrinkage for covariance estimation in the spiked model is considered in [1]. One may also use similar techniques to derive optimal shrinkers for loss functions other than Frobenius loss; see [3, 4, 1].

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